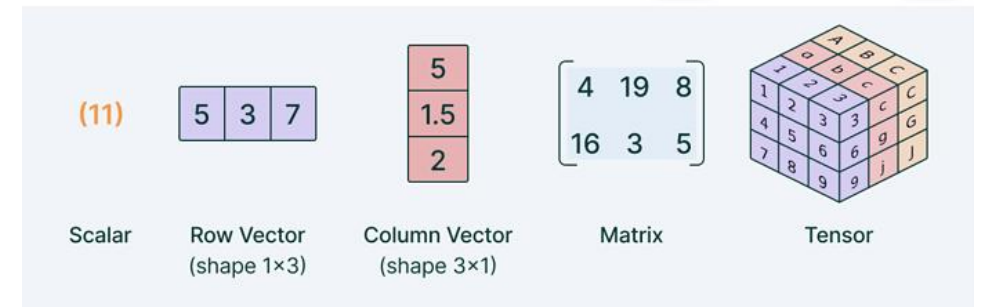


Tensor Operations: Moving Beyond NumPy

- Tensors are the fundamental data structure in deep learning, similar to NumPy arrays but with specialized capabilities.
- Deep learning frameworks allow for efficient management of memory between the **CPU** and **GPU (CUDA)**.
- Moving computations to the GPU provides the massive parallelization needed for training large models.



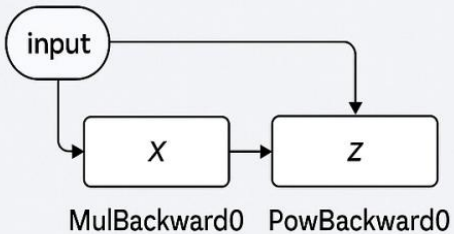
Autograd: The Engine of Learning

Understanding PyTorch Autograd vs AOTAutograd

Dynamic Gradient Graph, Compiler-Friendly Execution

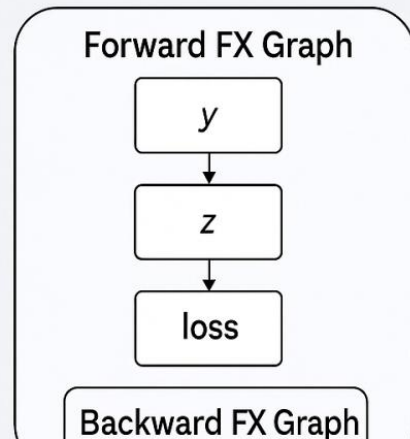
Autograd

Dynamic Gradient Graph



AOTAutograd

Static FX Graphs, Compiler-Friendly Exec



- **PyTorch** is our primary framework for tracking mathematical operations.
- **Autograd** automatically calculates the gradients (derivatives) required for backpropagation.
- By tracking every operation, the framework can automatically calculate how to update weights without manual calculus.

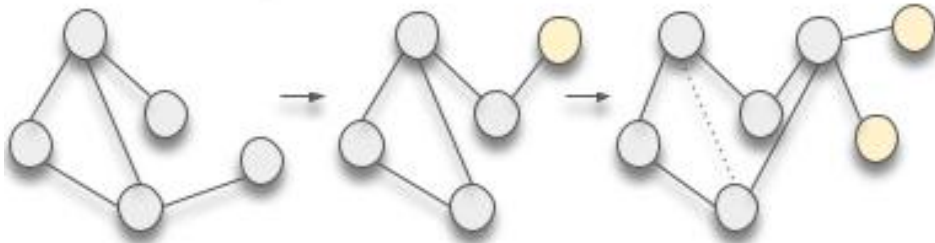
Computational Graphs: Static vs. Dynamic

Static Graph



Graph at time 1

Dynamic Graph



Graph at time 1

Graph at time 2

Graph at time 3

- A computational graph is a directed graph where nodes are operations and edges are tensors.
- **Static Graphs:** The graph is defined and compiled once (common in older TensorFlow).
- **Dynamic Graphs:** PyTorch builds the graph "on the fly" during execution, making debugging much easier and logic more flexible.